

Fantasy Anime League Kalman Filter Architecture

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September 2026

1 Introduction

Kalman Filters are used in GPS trackers, radars, and rockets. It's one of the reliable forecasting methods you can use when you don't have historical data for a set of parameters. It benefits greatly when you can easily measure data in short intervals of time. Kalman Filters are considered optimal because its estimates are unbiased and the Kalman gain (which we will discuss later) is chosen to minimize variance, hence minimal MSE.

The Random Forest model that predicts the optimal line-up for my FAL roster spits out both a predicted value and an uncertainty. A covariance matrix can be assembled from all possible metrics in a sample of 73 anime. With reasonable assumptions, a noise matrix and a transition state matrix can be set up. Hence, the Kalman Filter that was once used in rockets is a perfect fit for the wonderful and groundbreaking task of forecasting anime success.

2 Theory

We will start with an assumption: the 73 Fall 2026 anime metrics will have a distribution close to normal. This allows us to use z-score (which is the output of the neural network) as a proper metric. We will label the anime with a superscript such as x^i , but for simplicity purposes, let's remove the superscript and focus on one anime. Let's start with some initialization:

$$\hat{\mathbf{x}}_{0,0} = \mathbf{z}_0 = \begin{bmatrix} s_0 \\ w_0 \\ f_0 \\ d_0 \\ p_0 \end{bmatrix}$$

where p_0 is a variable for forum posts. Note that $\mathbf{z}_0$ is predicted from a Random Forest model. Note that the first subscript refers to the time of the system, and the second subscript is the time where the estimate was made. Also note they

are z-scores, we assume that they won't follow any trends—just slight to moderate noise.

Let $\mathbf{R}_0$ be our covariance matrix, which is derived from the means of the predicted metrics of all 72 anime. Making the first estimate involves the equation

$$\hat{\mathbf{x}}_{1,0} = \mathbf{F}\hat{\mathbf{x}}_{0,0}$$

where $\mathbf{F}$ is called the state transition matrix. There will be a section on my justification of the choice of $\mathbf{F}$.

The uncertainty of the current state $\mathbf{P}_{0,0}$ can be represented by the covariance matrix:

$$\mathbf{P}_{0,0} = \mathbf{R}_0$$

With this, we have a covariance extrapolation equation:

$$\mathbf{P}_{n+1,n} = \mathbf{F}\mathbf{P}_{n,n}\mathbf{F}^T + \mathbf{Q}$$

where $\mathbf{Q}$ is our process noise. Just like $\mathbf{F}$, there will be a section on how I choose $\mathbf{Q}$.

Now suppose Week 1 has officially started, and everyone now has points depending on their roster. We now have our first set of data, and of course, each metric will be a z-score against other Fall 2026 anime, namely $\mathbf{z}_1$. We will now focus on the covariance matrix of the 72 anime since we have that data now. Of course, this means that it'll be a lot noisier. This is where the power of the Kalman filter shows: how do we balance our "trust" towards the current measurement $\mathbf{z}_1$, the current covariance matrix $\mathbf{P}_{1,1} = \mathbf{R}_1$, the prediction $\hat{\mathbf{x}}_{1,0}$, and the predicted covariance matrix $\mathbf{P}_{1,0}$? We introduce the state update equation:

$$\hat{\mathbf{x}}_{1,1} = \hat{\mathbf{x}}_{1,0} + \mathbf{K}_1(\mathbf{z}_1 - \mathbf{H}\hat{\mathbf{x}}_{1,0})$$

where $\mathbf{K}_1$, the Kalman gain, can be calculated by

$$\mathbf{K}_n = \mathbf{P}_{n,n-1}\mathbf{H}^T (\mathbf{H}\mathbf{P}_{n,n-1}\mathbf{H}^T + \mathbf{R}_n)^{-1}$$

and $\mathbf{H}$ is the observation matrix that we will choose and justify in another section.

3 Supplementary Filters

Two more Kalman filters will be set up for the forecasting phase: one for the raw WC counts (for aging purposes) and one for the point distribution of metrics. The reason why the latter has to be done is because the main metrics are based off of z-scores, so in order to predict final scores, the individual z-scores are fed to a heuristic formula, and the coefficients for the formula depend on the point

distribution. Note that for the point distribution filter (sometimes I'll call it the criteria filter), each prediction will be normalized to have a total sum of 0.90. This is so that the code can make sure the heuristic formula doesn't make the predictions blow up.

4 Matrix Selection

Due to our models being simple, we will assume that our observation matrix $\mathbf{H}$ is the identity matrix for all three filters.

Main filter: since all metrics are assumed to be somewhat constant + slight to moderate noise, we will assign $\mathbf{F} = I_5$. Then we have $\mathbf{Q} = \text{diag}(\sigma_s, \sigma_w, \sigma_f, \sigma_d, \sigma_p)$, where each σ is the RMSE of each metric from the model prediction, scaled down by the standard deviation for each metric prediction.

Criteria filter: Same $\mathbf{F}$ as above, but the σ 's in the diagonal of $\mathbf{Q}$ will be chosen by eyeball estimation.

WC filter: We assume that raw WC counts have a local linear trend, hence we choose an $\mathbf{F}$ matrix of

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Additionally, for a random acceleration parameter σ , we assign a $\mathbf{Q}$ matrix of

$$\sigma^2 \begin{bmatrix} 1/4 & 1/2 \\ 1/2 & 1 \end{bmatrix}$$

5 Implementation

The file `update_json.py` has a `create_json` function that creates a fresh JSON. You don't have to worry about the incomplete content inside of it, the `update_json` function will handle that. When calling `update_json`, make sure to zero-index the day. Here are detailed instructions on how to do your daily updates:

1. Call `update_json` with the correct day number and overwrite the file
2. If it's not scoring week, copy paste the last day in `criteria.json`. If it is scoring week, update accordingly.
3. Load both JSONs in `forecasting.py` and run the entire pipeline on it.
4. Manually enter the predicted scores to a new day key in your predictions JSON file.
5. Load both predictions and `current_criteria` JSON files on `forecast_graphing.py`
6. Perform bench swaps and aces if necessary. I recommend doing them the day before scoring day.

6 Results

TBD.